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INFORMATICS PLATFORM

Deploying Webhooks in Customer-Hosted Environments Guide

Some customers of Sapio will have a local instance of the Sapio platform. Customer-hosted environments of Sapio (On-Premise, or formerly known as Sapio Platinum) provide a layer of local security and may be used instead of Sapio SaaS solutions. This article reviews how to deploy webhooks in customer-hosted environments of Sapio.

Within customer-hosted environments, Sapio On-Premise users may need, or wish, to use Webhooks. Webhooks are self-contained docker containers that run python code to achieve specific functions, primarily working as an interface layer between Sapio and external systems in response to an event trigger. When triggered to run, Webhooks communicate with the intended application or external system over HTTPS, or any other protocol deemed appropriate.

Sapio provides webhooks as docker containers to ensure a consistent execution environment, as well as facilitate joint development between Sapio and our customers.

This article describes the resource requirements for deploying webhooks in customer-hosted environments, considerations for running the webhook/docker, deployment options for architecture accessibility, and example connections for simple or high throughput/availability deployments.

Read the entire article or skip to your section of interest.

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Resource requirements

The baseline requirements for a webhook container are...

- 2CPU
- 4GB ram
- Considerations for overheads if the container runs on Linux or Windows due to other required software, such as monitoring and security agents in your corporate build

Considerations for Running the Webhook/Docker

When executing a docker run, users run the webhook container on the host machine. The host machine may be a local laptop or desktop for development, or a remote server. Processes in the docker container run in isolation, having their own file system, networking, and its own isolated process tree separate from the host. Users are free to run docker in whichever way best fits their infrastructure.

Note : Sapio uses AWS apprunner, but Sapio will be moving to AWS ECS to enhance performance

If you want to run docker on your own hardware on...

- **Linux and Windows :** we suggest running the container on a standalone server
- **Something else (like Kubernetes) :** we suggest this if you already have another system that can run docker containers set up, provided that the network level requirements your solution can be satisfied

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Deployment Options for Architecture Accessibility

Webhooks will need to be accessible from the application server and will also need to be able to call back to the application server, all over HTTPS. If you are running in a secure environment, this communication should take place within your own network.

Tip : The communication does not need to take place within your user network provided that the connection is secure

Users will generally need the following architecture depending on the nature of their runs and requirements...

- **If running general throughput or average availability requirements :** a simple deployment with one webhook container running for every webhook function that is required

- **If running high throughput or high availability requirements :** a high throughput/availability deployment with potentially more than one webhook container running for some webhook functions

Tip : A load balancer can be used If running multiple webhook instances. Ensure that any load balancers being used support the correct connection type. For example, an HTTP load balancer may not support native HL7 MLLP connections.

Example Connections

The following example connections can be used to deploy webhooks in customer-hosted environments

- Simple Deployment
- HighThroughput/Availability Deployment

Simple Deployment

This solution may be used if users do not need multiple instances to cater for throughput requirements and do not have any high-availability configuration for the application server.

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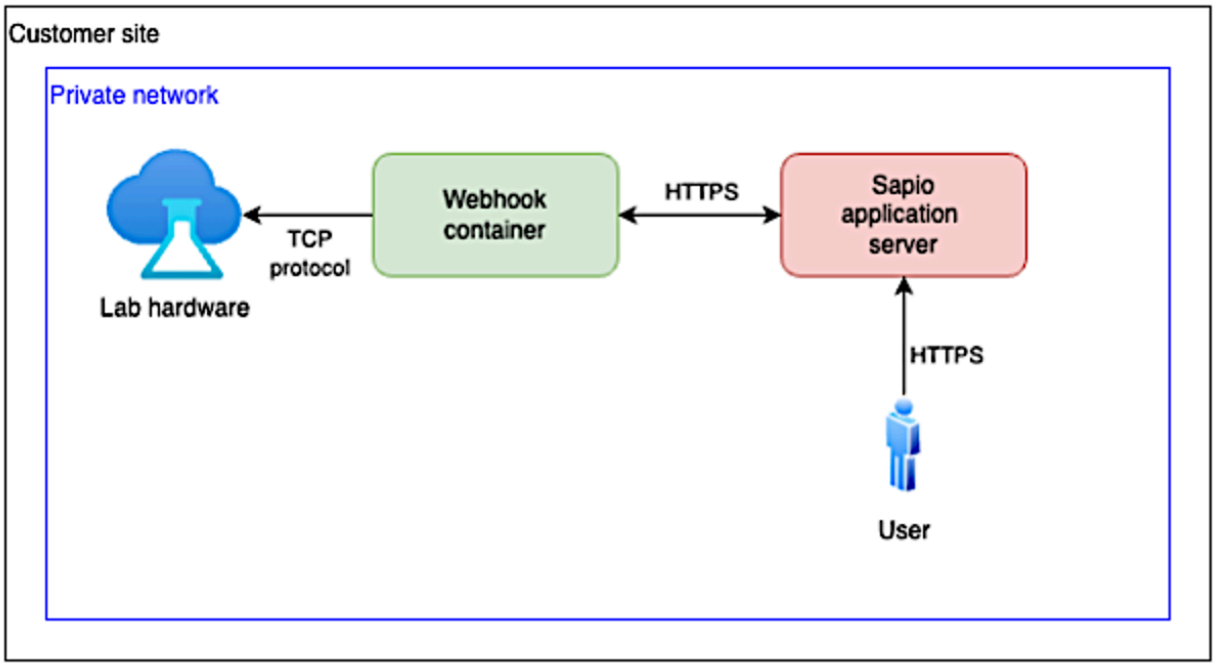
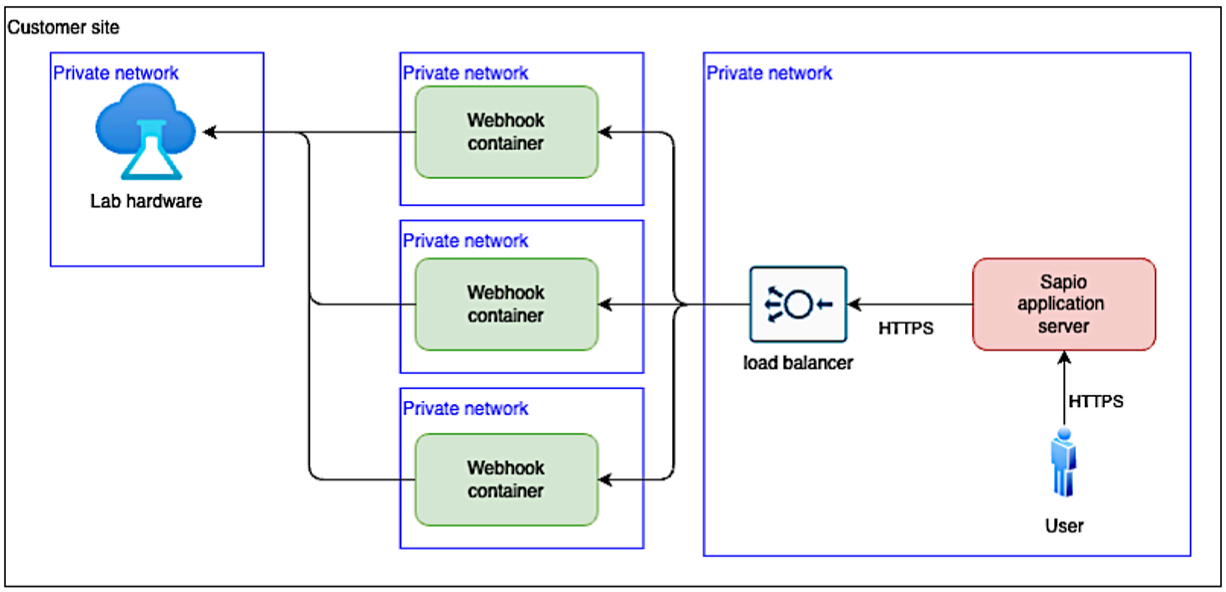


High Throughput/Availability Deployment

This solution may be used if users have high throughput or high availability requirements.

Note : No application high availability configuration is shown here for simplicity



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